



MACHINE LEARNING PROJECT

Topic

FEATURE EXTRACTION USING IMAGE DATA HCMUT – VNU-HCM

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1 Introduction

Automatically extracting meaningful representations from raw image data is a fundamental challenge in computer vision, where inputs are typically high-dimensional, complex, and subject to significant spatial variations (e.g., lighting, scale, and viewpoint shifts). Traditional hand-crafted feature extraction methods, such as SIFT or HOG, often fail to capture deep semantic hierarchies due to their reliance on rigid, low-level visual patterns.

In this project, we present a deep learning-based pipeline for image feature extraction and classification. Our approach leverages the hierarchical learning capabilities of Convolutional Neural Networks (specifically the ResNet18 architecture) to automatically discover and extract robust, discriminative visual features, replacing manual engineering with an end-to-end supervised learning paradigm.

2 Problem Statement

Most traditional image analysis systems rely either on raw pixel comparisons or manually engineered feature descriptors (such as SIFT, SURF, or HOG). While these methods are effective for extracting basic local gradients, they lack the capacity to generalize across complex variations in scale, rotation, illumination, and background clutter. This rigid approach introduces significant limitations and fails to capture the deep semantic meaning of an image.

In practice, real-world visual data—such as the images comprising the CIFAR-10 dataset—consist of unstructured, high-dimensional arrays of pixels. A raw image is merely a dense grid of RGB values, yet it simultaneously conveys complex, high-level semantic information (e.g., distinguishing a "truck" from an "automobile"). Such visual representations are instantly intuitive for human perception but highly non-linear and difficult for classical algorithms to process automatically due to the well-known "semantic gap."

The problem addressed in this project is to bridge the gap between low-level raw pixel data and high-level structured semantic understanding. Specifically, given a raw, low-resolution (32×32) colored image, the deep learning pipeline aims to automatically extract hierarchical representations and infer the following components:

- the low-level spatial patterns (e.g., edges, corners, and local textures)
- the high-level semantic parts representing the nature of the object
- the corresponding object category among a predefined set of classes

In particular, this visual recognition and feature extraction task is formulated as a supervised multi-class image classification problem, utilizing a deep Convolutional Neural Network (CNN) architecture to map raw image matrices directly to discrete class probabilities.

3 Motivation and Objectives

3.1 Motivation

The motivation for this project is strongly driven by the historical limitations and the manual overhead associated with traditional computer vision workflows. In classical image analysis, researchers and engineers were required to design rigid, hand-crafted feature descriptors (such as SIFT or HOG) to identify corners, edges, and textures. While structurally rigorous, this process involves significant human intervention, trial-and-error, and domain expertise just to extract basic representations from a single image.

In contrast, human visual perception operates in a much more fluid and intuitive manner. We instantly recognize complex objects from raw visual stimuli without explicitly calculating geometric properties or lighting conditions. Raw image data—though unstructured and highly variable—inherently contains all the necessary semantic information needed for classification, yet classical models fail to leverage this without extensive manual preprocessing.

This observation motivates the exploration of Convolutional Neural Networks (CNNs) as an automated feature extraction mechanism. By allowing the network to ingest raw pixel data directly, the system aims to eliminate the manual feature engineering overhead. Through hierarchical learning, the model can automatically

discover and extract structured visual features—from simple low-level edges to complex high-level object parts—and accurately classify them into discrete categories.

3.2 Objectives

The main objectives of this project are:

- To design and implement an end-to-end deep learning pipeline for automated image feature extraction and classification.
- To utilize the ResNet18 architecture to effectively capture hierarchical spatial patterns and overcome the vanishing gradient problem in deep networks.
- To explore the impact of data preprocessing and augmentation techniques on improving the model’s ability to generalize across visual variations.
- To experiment with deep learning optimization strategies (such as the Adam optimizer and Cross-Entropy loss) for multi-class image categorization.
- To establish a robust performance baseline on the CIFAR-10 dataset for future integration of more advanced architectures, such as Vision Transformers (ViTs), for better semantic understanding.

4 Related Work

Historically, image classification and feature extraction systems mainly relied on manually engineered pipelines and structured feature descriptors, which limited their ability to generalize across complex, highly variable raw visual data.

The figure conceptually illustrates the traditional design for visual recognition tasks. If a researcher wants to classify a single image, they typically have to go through multiple disparate steps: applying image filters, mathematically extracting local gradients or keypoints, constructing visual feature vectors, and finally feeding them into a separate classifier (such as an SVM) to output a prediction. In contrast, our proposed deep learning approach offers a streamlined, end-to-end process with raw pixel matrices as the single input.

In Computer Vision, classical approaches based on hand-crafted features (like HOG or SIFT) and classifiers such as Support Vector Machines or Random Forests have been widely used for simple object detection tasks due to their efficiency, low computational overhead, and mathematical interpretability.

Recent transformer-based models, including Vision Transformers (ViTs), have achieved state-of-the-art results in visual semantic understanding. However, such models often require massive datasets and significantly higher computational costs. This work focuses on Convolutional Neural Networks (specifically the ResNet18 architecture) as a highly robust, efficient deep learning baseline for feature extraction, while ViT-based approaches are considered in our future works.

4.1 Solution Approaches Overview

Existing approaches to image feature extraction and categorization can be broadly categorized into three groups. The first group consists of hand-crafted local descriptors (e.g., SIFT, SURF, HOG), which rely on manually designed mathematical operations to extract gradient information with high spatial precision but limited semantic generalization capability.

The second group employs mid-level representation models, such as the Bag of Visual Words (BoVW), to cluster local features and estimate category likelihoods based on visual vocabulary frequency. While more robust than raw local features, these models still heavily depend on the initial hand-crafted extraction step and struggle with high-level semantic ambiguity.

The third group includes deep representation learning approaches, where raw images are processed through multiple layers of artificial neurons, automatically mapping spatial pixels to dense feature vectors and directly to categories using fully connected classifiers. This project adopts this third strategy—specifically utilizing a deep CNN with residual connections—to bypass manual feature engineering entirely, balancing robust feature extraction, generalizability, and computational efficiency.

5 Dataset Description

To evaluate the proposed feature extraction and classification pipeline, we utilize the **Caltech-256** dataset, a widely recognized and highly challenging benchmark in the computer vision community. Sourced from Kaggle and originally introduced by researchers at the California Institute of Technology, the dataset contains a total of 30,607 real-world images spanning 256 distinct object categories, alongside an additional "clutter" category designed to test background rejection capabilities.

5.1 Dataset Characteristics

Unlike simpler datasets (such as MNIST or CIFAR-10), Caltech-256 introduces significant challenges that perfectly align with our problem statement regarding unstructured visual data:

- **High Intra-class Variance:** Objects within the same category appear in diverse poses, lighting conditions, and scales.
- **Background Clutter:** Unlike cropped datasets, objects in Caltech-256 are often embedded in complex, noisy backgrounds, requiring the feature extractor to effectively separate the foreground subject from irrelevant context.
- **Varying Resolutions:** The raw images do not have a uniform size or aspect ratio, reflecting the true nature of unconstrained real-world photography.

5.2 Data Distribution and Preparation Requirements

The dataset guarantees a minimum of 80 images per category, presenting a realistic scenario of mildly imbalanced data distribution. Because our deep learning architecture (ResNet18) requires structured, fixed-size tensors as input, the raw Caltech-256 images cannot be processed directly. This necessitates a robust data preprocessing pipeline—including resizing all images to a standardized resolution (e.g., 224×224 pixels), applying data augmentation techniques to prevent overfitting, and performing mathematical normalization to stabilize the gradient descent process during training.

5.3 Image Size Distribution and Structural Heterogeneity

To further illustrate the structural heterogeneity of the Caltech-256 dataset, Figure 1 presents the distribution of raw image dimensions (width versus height) across the entire dataset.

As shown in the scatter plot, the raw images exhibit extreme spatial variance. While there is a dense cluster of images around the mean dimensions of approximately 372×326 pixels, the spatial bounds are exceptionally wide. The dataset contains extremely low-resolution images (as small as 75×75 pixels) alongside ultra-high-resolution photographs (reaching up to 3504×3468 pixels). Furthermore, the wide spread of the data points away from a perfect diagonal line indicates significant inconsistencies in aspect ratios.

This massive variance poses a direct challenge for our deep learning pipeline. Convolutional Neural Networks like ResNet18 strictly require standardized, fixed-size input tensors (typically 224×224). The extreme dimensional disparity means that naively resizing these images could lead to severe consequences:

- **Aspect Ratio Distortion:** Forcing non-square images into a square tensor will stretch or squish the objects, potentially altering their spatial semantic meaning.
- **Information Loss and Artifacts:** Downsampling a 3500×3500 image to 224×224 discards massive amounts of fine-grained detail, while upsampling a 75×75 image introduces blurring and interpolation artifacts.

Consequently, this dimensional analysis justifies the critical need for a robust data preprocessing and augmentation pipeline before feature extraction can occur.

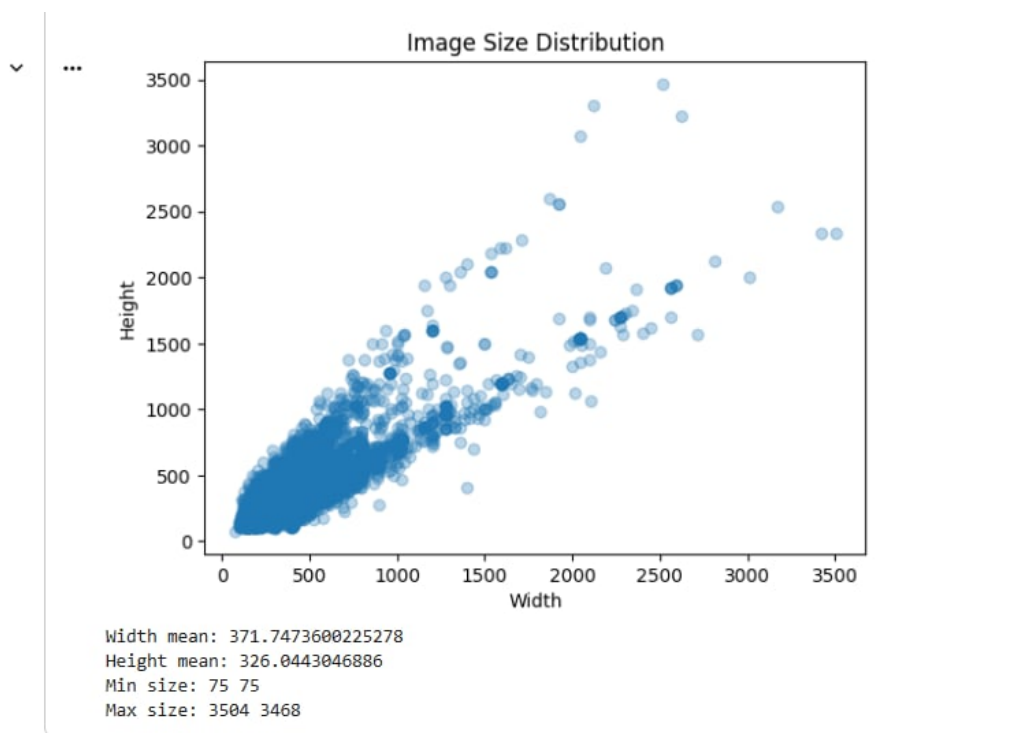


Figure 1: Scatter plot of image size distribution in the Caltech-256 dataset.